

Thermal management of metal hydride hydrogen storage tank coupled with proton exchange membrane fuel cells

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ABSTRACT

Metal hydride (MH), considered a promising hydrogen storage material, has received wide attention. As an efficient device for hydrogen energy utilization, a proton exchange membrane fuel cell (PEMFC) generates power and releases waste heat. MH tank can supply hydrogen for PEMFC, and the waste heat produced by PEMFC can be applied to heat the MH bed for dehydrogenation reaction. Thermal management between MH tank and PEMFC has been studied in this work. A lumped parameter model for the PEMFC-MH system has been developed and validated. The performance of the PEMFC-MH system has been predicted, the results show that it is feasible to use waste heat produced by PEMFC for heating the MH bed. The effect of the valve's opening degree on system performance has been investigated. The performance of the PEMFC-MH system with various output powers has been studied. The hydrogen supply rate from the MH bed to PEMFC can be adjusted based on different power requirements. The valve's opening degree can be considered an effective control measure in thermal management to improve the energy efficiency of the PEMFC-MH system.

1. Introduction

The shipping industry plays an important role in the national economy. However, traditional diesel-powered ships consume a lot of fossil energy with a large amount of CO₂ emitted. Various governments and enterprises actively seek solutions to reduce CO₂ emissions from shipping in the face of global climate change [1]. The hydrogen-powered ships have received wide attention [2,3]. Considering the small power requirement and short route for inland vessels, the inland river may be one of the main application scenarios of hydrogen-powered ships [4]. Of course, effective hydrogen storage methods for the shipping industry are underway, which is essential to wide applications of hydrogen energy.

As a kind of effective hydrogen storage material, metal hydride (MH) has been successfully applied to the submarine [5] and shows

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the possibility of its application in the light vehicle [6] and the waterway transportation field [2]. Considering the relatively low operating pressure of hydrogen storage systems based on MHs, it may cause the refueling cost and the hydrogen price to decrease [7], which is good for promoting hydrogen applications. As a kind of common MH, LaNi_5 hydrides have some advantages, such as low operating temperature and platform pressure, and have received extensive research [8–12]. Compared with some high-temperature MHs, such as Mg-based hydride [13], its gravimetric hydrogen density is relatively small. However, compared with the gravimetric energy density of the Li-ion battery systems in the ships (about 111 Wh/kg [14]), MHs present better energy storage performance when the stored hydrogen is converted into electric energy by PEMFC.

In general, the hydriding/dehydriding reaction is accompanied by obvious exothermic/endothermic phenomenon, which leads to significant thermal effect problems and affects system performance. Various heat exchangers, such as metal foams [8,9] and cooling tubes [11], have been applied in MH reactors to mitigate the thermal effects of the chemical reaction processes. Metal foam with relatively large thermal conductivity can improve the heat transfer performance of MH beds by increasing its effective thermal conductivity [8]. Phase change material (PCM) jacket can be used as a passive heat exchanger, which absorbs heat during hydriding and releases it during dehydriding in the MH system [15–17]. Considering the relatively low thermal conductivity of PCM, it can work with metal foam [15] or expanded graphite [16] and form composited PCM with higher thermal conductivity, which is proven to be feasible to improve the hydriding rate of MH systems. MH reactor with PCM has been applied to supply hydrogen for fuel cell applications [16,17]. A cooling tube or heat exchanger can also be used as the bridge to enhance heat transfer between the MH bed and the PCM jacket, which is good for improving the reaction rate and reusing the heat stored in PCM [18].

Thermal management between MH and other equipment has received wide attention. The feasibility of cooling the battery pack with the MH during dehydriding has been investigated by experiments for a hybrid electric vehicle [19]. In recent years, many researchers have been trying to reuse waste heat as the by-product of PEMFC, which makes it possible to supply heat for MH beds [20–22]. Proton exchange membrane fuel cells can be considered an efficient device utilization of hydrogen energy, where there are complicated electrochemical and thermal processes [23,24]. Based on the experimental data in Ref. [25], the thermal management of the PEMFC stack for a vehicle has been investigated [26]. The heat produced by the PEMFC stack can be directly transferred to MH tank by using electric fan [22], which can be considered a simple and feasible way. However, the hot air from the fan is used to heat the outer wall of the MH tank. Normally, directly heating the MH bed rather than the wall of the tank will show better results.

Some special devices based on heat exchanger material have been designed and developed as the bridge for strengthening the heat transfer between the PEMFC and the MH bed. As an efficient heat exchanger, the heat pipe can be used as the thermal bridge between them, which can effectively improve the hydrogen supply rate of the hydrogen storage tank [27]. Better performance can be obtained when the heat pipe is covered by fins [28]. As for the PEMFC with relatively large output power, the liquid cooling mode is more popular than the air cooling mode. It will consume electric energy generated by the PEMFC stack when the cooling system works. Some waste heat released by PEMFC is transferred to the MH bed, then the energy consumption of the auxiliary cooling sub-system of PEMFC decreases, which leads to an increase in the system generation efficiency. As for a hydrogen-powered vehicle where the waste heat released by PEMFC is transferred to a MH tank through a fan and radiator, research results show that this integration equipment can save 10% fuel [29]. In the thermal management module of the PEMFC-MH system, the hydrogen storage tank and the heat exchanger can be connected in serial [30] or parallel [21,29] in the circulating water system. Generally, it is good for controlling the dehydriding rate by adjusting the valve opening for the parallel mode. The PEMFC-MH system was applied to the first Italian hydrogen propulsion ship [2], and its corresponding control method for satisfying various load demands plays an important role in the practical application process [31].

In our previous work [32], the dehydriding performance of MH tank has been studied based on the validated distributed numerical model, and the hydrogen fuel supply rate of this MH reactor for applications can be fitted as an equation about the pressure at the outlet and the temperature of heating water. Meanwhile, some lumped parameter models of MH reactors have been validated [8,9] and applied for hydrogen storage and purification [33]. This work has developed a lumped parameter model for the MH hydrogen storage tank coupled with PEMFC. The feasibilities of directly supplying hydrogen from the MH tank to PEMFC and using waste heat produced by PEMFC for heating MH bed during dehydriding have been evaluated, and the thermal management of this PEMFC-MH system has been studied under different output power conditions.

2. Mathematical model

The basic governing equations for the lumped parameter model of the PEMFC-MH system have been introduced in this section. This lumped parameter model mainly consists of the PEMFC stack, the MH hydrogen storage tank and the circulating water sub-system used for the thermal management between the PEMFC stack and the MH tank. The MH tank can be considered a hydrogen source to supply high-purity fuel for the PEMFC system. The PEMFC stack generates electricity by producing much heat by consuming the charged hydrogen. The thermal management sub-system is designed to transfer adequate waste heat from the PEMFC to the MH tank during dehydriding and transfer the rest of heat to the environment by the heat exchanger to maintain that the PEMFC system works at the proper temperature. This method is not only good for MH tank to supply enough heat in the dehydriding process but also good for PEMFC system to decrease the energy consumption of the cooling device and improve the system efficiency.

2.1. PEMFC

The output voltage of a single fuel cell V_{cell} [V] is relative to the Nernst open circuit voltage E_{Nernst} [V] and three voltage losses, including the activation voltage loss V_{act} [V], the ohmic voltage loss V_{ohm} [V] and the concentration voltage loss V_{com} [V], which can be shown as,

$$V_{\text{cell}} = E_{\text{nernst}} - V_{\text{act}} - V_{\text{ohm}} - V_{\text{com}} \quad (1)$$

The Nernst open circuit voltage E_{nernst} can be written as [26],

$$E_{\text{nernst}} = 1.229 - 8.5 \times 10^{-4}(T_{\text{FC}} - 298.15) + 4.308 \times 10^{-5} T_{\text{FC}} \ln(p_{\text{H}_2} p_{\text{O}_2}^{0.5}) \quad (2)$$

where T_{FC} [K] means the temperature of PEMFC, p_{H_2} [bar] and p_{O_2} [bar] are the partial pressure of hydrogen in the anode and the partial pressure of oxygen in the cathode, respectively. The partial pressure of hydrogen p_{H_2} is relative to the anode pressure p_{an} [bar], the current density i_{FC} [A/cm²], T_{FC} and the saturation pressure of water vapor $p_{\text{H}_2\text{O}}^{\text{sat}}$ [bar], as presented in Ref. [23],

$$p_{\text{H}_2} = 0.5 \times \left[\frac{p_{\text{an}}}{\exp\left(\frac{1.653 i_{\text{FC}}}{T_{\text{FC}}^{1.334}}\right)} - p_{\text{H}_2\text{O}}^{\text{sat}} \right] \quad (3)$$

The partial pressure of oxygen can be written as [23],

$$p_{\text{O}_2} = \frac{p_{\text{ca}}}{\exp\left(\frac{4.192 i_{\text{FC}}}{T_{\text{FC}}^{1.334}}\right)} - p_{\text{H}_2\text{O}}^{\text{sat}} \quad (4)$$

The saturation pressure of water vapor $p_{\text{H}_2\text{O}}^{\text{sat}}$ is relative to the temperature of the fuel cell T_{FC} , which is shown in Eq. (5) [23]. The inlet gas pressure of the anode and the cathode is set as 1 bar [25,26].

$$\log_{10}(p_{\text{H}_2\text{O}}^{\text{sat}}) = -2.18 + 2.95 \times 10^{-2}(T_{\text{FC}} - 273.15) - 9.19 \times 10^{-5}(T_{\text{FC}} - 273.15)^2 + 1.44 \times 10^{-7}(T_{\text{FC}} - 273.15)^3 \quad (5)$$

The activation voltage loss V_{act} can be presented as [23],

$$V_{\text{act}} = - \left(-0.948 + 0.00304 T_{\text{FC}} + 7.6 \times 10^{-5} T_{\text{FC}} \ln C_{\text{O}_2} - 1.93 \times 10^{-4} T_{\text{FC}} \ln I_{\text{cell}} \right) \quad (6)$$

where I_{cell} [A] means current and the oxygen concentration C_{O_2} [mol/cm³] is equal to $\frac{p_{\text{O}_2} \exp(498/T_{\text{FC}})}{5.08 \times 10^6}$ [23].

The ohmic voltage loss V_{ohm} is relative to the current density i_{FC} , the membrane resistance r_{m} [Ωcm] and the thickness of PEM τ_{m} [cm] when the contact resistance is not taken into consideration [26], which can be expressed as,

$$V_{\text{ohm}} = i_{\text{FC}} r_{\text{m}} \tau_{\text{m}} \quad (7)$$

The membrane resistance r_{m} is presented as [24,25],

$$r_{\text{m}} = \frac{181.6 \left[1 + 0.03 i_{\text{FC}} + 0.062 \left(\frac{T_{\text{FC}}}{303} \right)^2 i_{\text{FC}}^{2.5} \right]}{(\lambda - 0.634 - 3 i_{\text{FC}}) \exp \left[4.18 \left(\frac{T_{\text{FC}} - 303}{T_{\text{FC}}} \right) \right]} \quad (8)$$

where the membrane humidity λ is set as 7.25 in this work.

The concentration voltage loss V_{com} is relative to the current density i_{FC} . The maximum current density is set as 1.6 A/cm², and the empirical parameter B [V] in Eq. (9) is set as -0.016 [23,26].

$$V_{\text{com}} = -B \ln \left(1 - \frac{i_{\text{FC}}}{1.6} \right) \quad (9)$$

The hydrogen released from the MH tank flows into the PEMFC stack. The hydrogen stoichiometry φ is defined as the ratio between the amount of hydrogen that flowed into the fuel cell and the amount of hydrogen that reacted in the fuel cell, which is set to 1 in this work. The relationship between the mass flow rate of hydrogen $\dot{m}_{\text{H}_2}$ and the current of a single PEM fuel cell I_{FC} can be presented as follows,

$$\dot{m}_{\text{H}_2} = \frac{I_{\text{FC}} \varphi M_{\text{H}_2} N_{\text{FC}}}{2F} \quad (10)$$

where the Faraday constant $F = 96485$ C/mol and N_{FC} is the number of single fuel cell.

The PEMFC is a device that generates power from hydrogen by releasing a large amount of waste heat. It is disadvantageous for the PEMFC to work at a relatively high temperature. In general, the heat produced by PEMFC must be removed by the heat exchanger or the cooling device to the environment.

The heat generation of PEMFC Q_{FC} [W] can be written as [26],

$$Q_{\text{FC}} = N_{\text{cell}} (E_{\text{nernst}} - V_{\text{cell}}) I_{\text{FC}} \quad (11)$$

2.2. Metal hydride hydrogen storage tank

The MH tank is used to supply hydrogen fuel for the PEMFC system. In general, the MH bed needs to be heated to obtain an

acceptable dehydrating rate. Considering this work focuses on studying the hydrogen supply performance of the MH tank for the PEMFC, the governing equations during dehydrating are introduced. The mathematical model during hydriding can be found in our previous works [12,33].

The mass conservation equations of the MH bed and the gaseous hydrogen in the tank can be expressed as Eq. (12) and Eq. (13) [9, 33],

$$\frac{dm_{MH}}{dt} = m_{MH} C_d e^{-E_d/RT} \left(\frac{p_{H_2} - p_{eq}}{p_{eq}} \right) \quad (12)$$

$$\frac{dm_{H_2}}{dt} = \dot{S}_{H_2} - \dot{m}_{H_2} \quad (13)$$

The amount of gaseous hydrogen in the MH tank is related to the mass source term for hydrogen $\dot{S}_{H_2}$ [g/s] and the mass flow rate $\dot{m}_{H_2}$ [g/s] from the hydrogen storage tank into the PEMFC stack. The hydrogen mass source term $\dot{S}_{H_2}$ is related to the dehydrating rate of the MH bed and can be expressed as $\frac{dm_{MH}}{dt} \frac{M_{H_2} SC}{M_{MH}}$. The molecular weight of metal hydride M_{MH} is equal to 438 g/mol. The stoichiometric coefficient of hydrogen in the hydriding/dehydrating reaction SC is 3 in this work.

The pressure of hydrogen p_{H_2} [bar] can be calculated by the state equation for ideal gases, which is relative to the mass and the temperature of gaseous hydrogen. The equation of state for ideas gases can be written as $pV = nRT$ in a general form. As for special gases, such as gaseous hydrogen, it also can be presented as $p = \rho R_{H_2} T$. The relationship between the gas constant of hydrogen R_{H_2} and the universal gas constant R can be expressed as $R_{H_2} = R/M_{H_2}$. The molecular mass of hydrogen M_{H_2} is equal to 2 g/mol.

The gaseous hydrogen mass in the tank can be calculated from the dehydrating rate of MH bed and the mass flow rate of hydrogen supplied for PEMFC. The temperature of gaseous hydrogen is assumed to be the same as that of MH bed T_{MH} [K].

The equilibrium pressure p_{eq} [bar] is shown as [15],

$$\ln \left(\frac{p_{eq}}{p_0} \right) = -\frac{\Delta H}{RT_{MH}} + \frac{\Delta S}{R} \quad (14)$$

where the reference pressure p_0 is equal to 1 bar, the entropy of reaction ΔS is equal to 108 J/mol/K, and the enthalpy of reaction ΔH is equal to 30800 J/mol. The equilibrium pressure is reduced from the Van't Hoff equation, where the symbol R means the universal gas constant rather than the gas constant of hydrogen. The universal gas constant R is equal to 8.314 J/mol/K.

The energy conservation for the hydrogen storage tank can be expressed as [9,33],

$$(m_{H_2} c_{pH_2} + m_s c_{ps}) \frac{dT_{MH}}{dt} = A_{W_MH} U (T_{W_MH_in} - T_{MH}) + \Delta H \frac{dm_{MH}}{dt} \frac{SC}{M_{MH}} \quad (15)$$

where the term $A_{W_MH} U (T_{W_MH_in} - T_{MH})$ can be defined as the heat Q_1 [W] transferred from the heat exchanger into the MH tank. The symbol A_{W_MH} [m²] means the heat exchange area between the circulating water and the MH bed, and the wall of circulating water is ignored.

The heat transfer coefficient between the MH bed and the circulating water U [W/m²/K] can be calculated as follows,

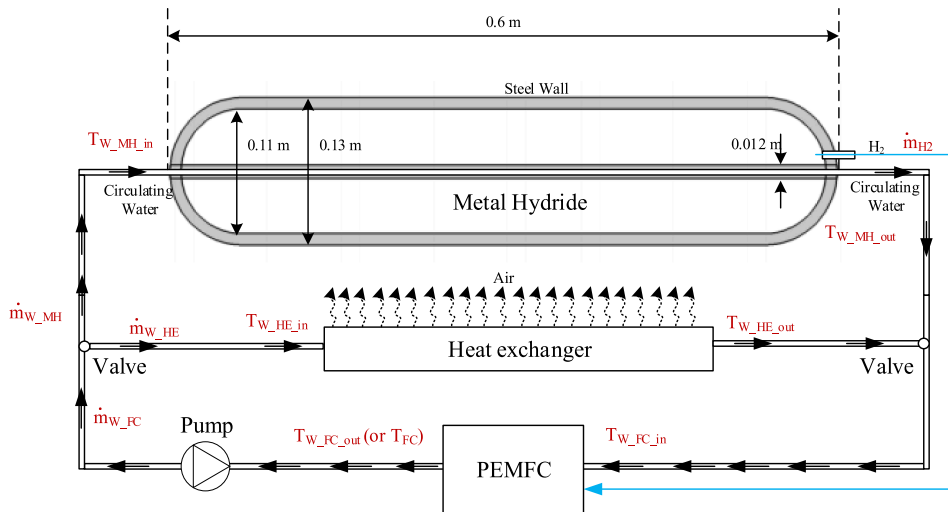


Fig. 1. Schematic design of metal hydride hydrogen storage tank coupled with proton exchange membrane fuel cells.

$$U = 1.86 \text{Re}_f^{1/3} \text{Pr}_f^{1/3} \left(\frac{d_3}{L} \right)^{1/3} \left(\frac{\mu_f}{\mu_w} \right)^{0.14} \lambda_{d_3} \quad (16)$$

where Re_f means the Reynolds number of fluid, Pr_f is the Prandtl number of fluid, and more details have been given in Ref. [34].

In addition, the hydrogen storage capacity of the MH bed wt [%] is defined as $m_H/m_s \times 100$ in this work, m_H [g] means the mass of hydrogen stored in the MH bed and m_s [g] is the mass of the MH bed. This parameter wt can be considered a reference to measure the amount of residual hydrogen stored in the reactor in the fuel supply process.

2.3. Thermal management between PEMFC and metal hydride tank

The thermal management sub-system plays an important role in maintaining the normal operation of the PEMFC-MH system. As for the PEMFC-MH system in the present work, a portion of the heat produced by the PEMFC stack can be used to heat the MH bed for quick hydrogen release. It provides a feasible way for the waste heat utilization of PEMFC. The heat exchanger transfers the rest waste heat from the circulating water to the environment.

The diagram of the PEMFC-MH system is shown in Fig. 1. The circulating water in the tube is used as the heat exchange medium among the PEMFC stack, the MH bed and the heat exchanger. The circulating water with a relatively high temperature flows out from the PEMFC stack, and flows to the MH tank and the heat exchanger through a three-way valve. Then, the circulating water with a relatively low temperature from these two devices flows into the PEMFC stack through another three-way valve. The mass flow rate of circulating water flowing through the PEMFC stack $\dot{m}_{W_FC}$ [kg/s] can be controlled by the pump.

The energy balance equation of the circulating water between the inlet and the outlet of the PEMFC stack section can be expressed as,

$$\dot{m}_{W_FC} c_{pW} T_{W_FC_out} = Q_{FC} + \dot{m}_{W_FC} c_{pW} T_{W_FC_in} \quad (17)$$

where $\dot{m}_{W_FC}$ means the mass flow rate of circulating water and is set to be constant in this work, $T_{W_FC_out}$ [K] and $T_{W_FC_in}$ [K] are the temperature of circulating water at the outlet and the inlet of the PEMFC stack, respectively. For simplification, $T_{W_FC_out}$ is set to be equal to the operation temperature of PEMFC T_{FC} . The specific heat capacity of circulating water c_{pW} is equal to 4185 J/kg/K. The heat generation of PEMFC Q_{FC} is relative to the hydrogen consumption rate, which can be considered the mass flow rate of hydrogen $\dot{m}_{H_2}$ supplied from the MH tank to the PEMFC stack.

The energy balance equation of the circulating water between the inlet and the outlet of the MH bed section can be written as,

$$\dot{m}_{W_MH} c_{pW} T_{W_MH_in} = Q_{MH} + \dot{m}_{W_MH} c_{pW} T_{W_MH_out} \quad (18)$$

where Q_{MH} means the heat transferred from the circulating water to the MH bed. The mass flow rate of circulating water flowing through the MH bed $\dot{m}_{W_MH}$ can be controlled by the opening percentage of the three-way valve a , which is equal to $a\dot{m}_{W_FC}$. The temperature of circulating water at the inlet of the MH bed $T_{W_MH_in}$ is simplified as T_{FC} .

As for the heat exchanger section, its energy balance equation can be shown as,

$$\dot{m}_{W_HE} c_{pW} T_{W_HE_in} = Q_{HE} + \dot{m}_{W_HE} c_{pW} T_{W_HE_out} \quad (19)$$

where $T_{W_HE_in} = T_{FC}$ and $\dot{m}_{W_HE} = (1 - a)\dot{m}_{W_FC}$. The symbol Q_{HE} [W] means the heat transferred from the circulating water to the environment through the heat exchanger.

The circulating water from the PEMFC flows partially to heat the MH bed and residually to be cooled by the heat exchanger, and then both water currents flow into another three-way valve and merge to the cooling tube in the PEMFC stack. The energy balance equation can be presented as,

$$\dot{m}_{W_FC} c_{pW} T_{W_FC_in} = a\dot{m}_{W_FC} c_{pW} T_{W_MH_out} + (1 - a)\dot{m}_{W_FC} c_{pW} T_{W_HE_out} \quad (20)$$

The temperature of circulating water at the inlet of the PEMFC stack $T_{W_FC_in}$ can be presented as $aT_{W_MH_out} + (1 - a)T_{W_HE_out}$.

3. Model parameters and validation

3.1. Model parameters

A lumped parameter model of the PEMFC stack is applied to the PEMFC-MH system. As a power generation device, the PEMFC produces electricity and releases waste heat as a by-product by consuming hydrogen. In order to meet the temperature limitation of proton exchange membranes, the operation temperature of the PEMFC stack is almost below 90 °C, which is dependent on the thermal management strategy of the PEMFC system. In this work, the operation temperature is set to be constant. The PEMFC stack consists of 45 single fuel cells. The active area of the single fuel cell is 50 cm² and its thickness of the membrane is 0.00508 cm, more information about the single cell can be found in Refs. [25,26].

As shown in Fig. 1, the hydrogen storage tank is equipped with a straight tube heat exchanger for heating the MH bed during dehydriding. The outer diameter of the tank d_1 is 0.13 m and its inner diameter d_2 is 0.11 m. The total length of the MH tank is 0.6 m, and the length of the cooling tube in the MH bed L is 0.58 m. The outer diameter of the cooling tube d_3 is 0.012 m, and the effect of the steel wall of the cooling tube on system performance is ignored in this lumped parameter model. The heat transfer area A_{W_MH} between the MH bed and the circulating water has been calculated and set as about 0.02187 m². The metal hydride LaNi_5H_6 is a kind of common

hydrogen storage material which can release hydrogen at room temperature. It also shows good performance of hydriding/dehydriding reaction kinetics. Of course, some other MHs with a relatively low operation temperature, such as TiFe hydrides, can be applied to the PEMFC-MH system in the future. The hydrogen storage tank is filled with this material, and the volume of the MH bed is about 70% of the total volume of this tank [21]. The change in the volume of the MH bed during dehydriding is neglected, and the porosity of the MH bed is assumed to be constant (0.5 [-]). The solid material is in the saturated state at the initial stage, and its density is equal to 7259 kg/m^3 [10,15]. There is about 12.95 kg of MH in the tank, which can be used to store about 177.4 g of H_2 . The initial temperature and the initial pressure are set as 323 K and 4.576 bar, respectively. More information on the parameters used in the simulation of the PEMFC-MH system is presented in Table 1.

3.2. Model validation

Based on the mathematical equations and the parameters above, the lumped parameter model of PEMFC can be developed on the software platform Matlab/Simulink. The flowchart of calculation for the PEMFC-MH system has been presented in Fig. 2. Based on the power requirement of load, the hydrogen supply rate from the MH tank to PEMFC is determined. Then, PEMFC can generate power for a load with much heat released. By setting the opening degree of the valve in the thermal management of the PEMFC-MH system, the waste heat can be reused for a dehydriding reaction. Its simulation results are compared with the data from other published works [25, 26], including the voltage and the thermal power of a single fuel cell at different operation temperatures (323 K and 353 K), as shown in Fig. 3. In general, the voltage decreases and the thermal power increases with the increment of the current density. The difference between the present work and others is acceptable, so this PEMFC model can be considered validated.

In addition, various lumped parameter models of hydrogen storage tanks filled with MH have been validated in our previous work [33]. The MH tank is used to supply hydrogen fuel for PEMFC in order to satisfy the requirement of loads. Some waste heat as the by-product of PEMFC is applied to heat the MH tank and improve the dehydriding rate through the circulating water system. Basic energy balance equations are used to describe the thermal management between the MH tank and the PEMFC system, whose correctness is self-evident. In general, the PEMFC-MH system can be considered validated based on the effectiveness of these sub-system models. This lumped parameter model can be used as a basic reference to estimate the feasibility of the PEMFC-MH system and find reasonable operating conditions.

4. Case studies

4.1. Performance prediction of PEMFC-MH system

The performance of the PEMFC-MH system has been predicted in order to evaluate the feasibility of heating the MH bed with the waste heat of PEMFC.

The hydrogen supply rate of the MH tank $\dot{m}_{\text{H}_2}$ is set as $2.25 \times 10^{-2} \text{ g/s}$ (about 15.17 L/min), which is assumed to be equal to the hydrogen consumption rate of PEMFC. The changes in hydrogen pressure between the MH bed and the PEMFC are not considered in this work. The hydrogen flowing into the PEMFC is assumed to be consumed completely. The operation temperature of PEMFC T_{FC} and the opening degree of the three-way valve a are set as 323 K and 0.3, respectively. The temperatures of circulating water at the inlet of the MH bed and heat exchanger are assumed to be equal to T_{FC} .

The simulation results have been presented in Fig. 4, including the hydrogen pressure in the MH tank, the average temperature of the MH bed, the hydrogen storage capacity of the MH bed, the heat power of the PEMFC system, the MH bed and the heat exchanger, the temperatures of circulation water at the inlet and the outlet of MH bed and the temperatures of circulation water at the inlet and the outlet of heat exchanger. As shown in Fig. 4a, the maximum pressure of hydrogen in the MH tank is about 3.5 bar (0.35 MPa). Considering the relatively low operating pressure of MHs, the max pressure in the MH reactor is much lower than the normal working pressure of the compressed hydrogen storage tank (35 MPa/70 MPa). We can use the low-cost metal tank for the MHs.

When the gaseous hydrogen flows into the PEMFC stack from the MH tank at a constant rate, the hydrogen storage capacity of the MH bed almost decreases linearly, and the hydrogen pressure in the tank decreases to below 1.1 bar after 2170 s. Under these conditions, it is necessary to equip a device to increase the pressure of hydrogen in order to meet the requirement for the inlet pressure at the anode.

Table 1
Parameters used in the simulation [9,10,15,25,26,33].

Parameter	Values	Parameter	Values
Active area of single fuel cell A , $[\text{cm}^2]$	50	Thickness of membrane, τ_m [cm]	0.00508
Number of fuel cells, N_{cell} [-]	45	Faraday constant, F [C/mol]	96485
Inlet gas pressure of the anode, p_{an} [bar]	1	Inlet gas pressure of the cathode, p_{ca} [bar]	1
Kinetics constant, C_d [1/s]	9.6	Enthalpy of reaction, ΔH [J/mol]	30800
Activation energy, E_d [J/mol]	16420	Entropy of reaction, ΔS [J/mol/K]	108
Molecular mass of hydrogen, M_{H_2} [g/mol]	2	Specific heat capacity of metal hydride, c_{ps} [J/g/K]	0.355
Molecular mass of hydrogen, M_{MH} [g/mol]	438	Specific heat capacity of hydrogen, c_{pH_2} [J/g/K]	14.3
Porosity of metal hydride bed, ϵ [-]	0.5	Reference pressure, p_0 [bar]	1
Hydrogen stoichiometry in PEMFC, φ [1]	1	Universal gas constant, R [J/mol/K]	8.314
Hydrogen stoichiometry in MH, SC [1]	3	Specific heat capacity of water, c_{pw} [J/kg/K]	4185
Mass flow rate of circulating water flowing through PEMFC, $\dot{m}_{\text{W_FC}}$ [kg/s]	0.02608	Heat exchange area between MH bed and circulating water, $A_{\text{W_MH}}$ [m^2]	0.02187

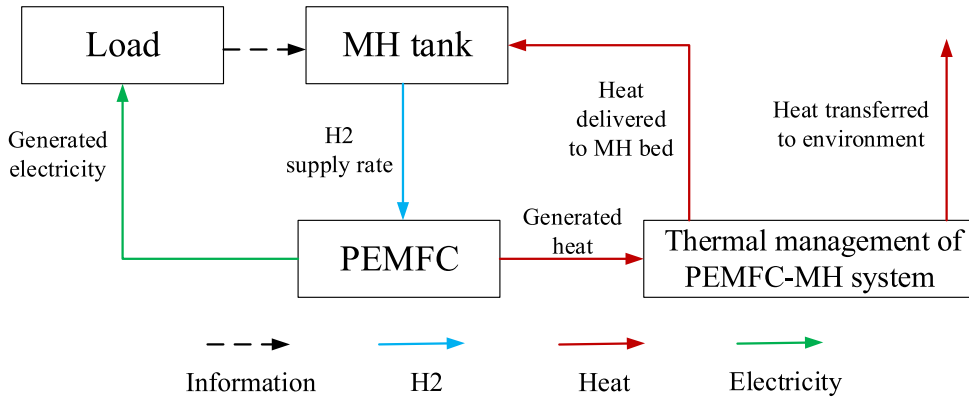
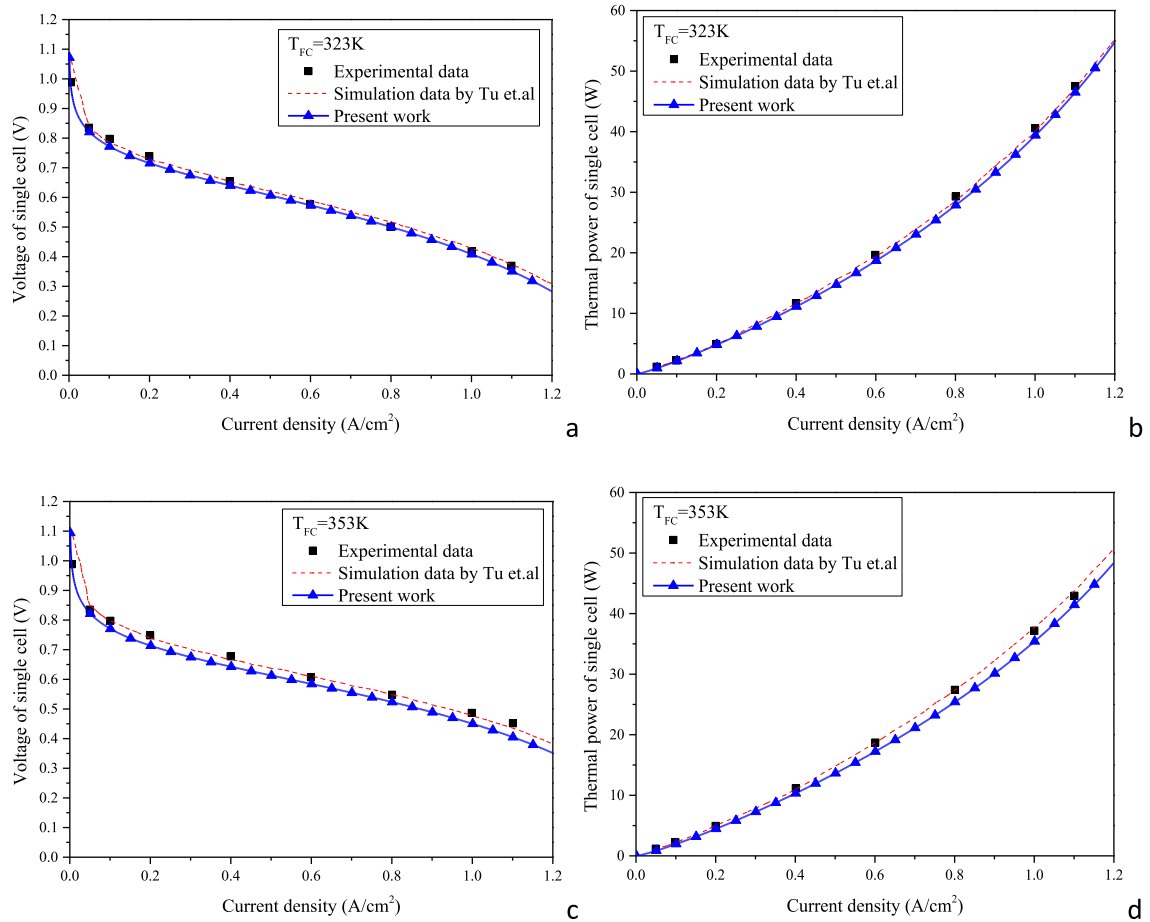


Fig. 2. Flowchart of calculation for PEMFC-MH system.

Fig. 3. Model validation for single proton exchange membrane fuel cell, including the output voltage (a, c) and thermal power (b, d) at $T_{FC} = 323\text{ K}$ (a, b) and $T_{FC} = 353\text{ K}$ (c, d).

Given the constant hydrogen supply rate for the PEMFC, the heat power Q_{FC} can be considered a fixed value. The heat generated by PEMFC is divided into two parts, one is used to heat the MH bed through the circulation water, and the other is transferred to the environment through the heat exchanger. The ratio of the amount of heat produced by PEMFC to that delivered to the MH bed can be written as Q_{MH}/Q_{FC} . The simulation results show that the value of the expression Q_{MH}/Q_{FC} is no more than 21%, which means the heat produced by PEMFC is enough for heating MH bed under ideal conditions.

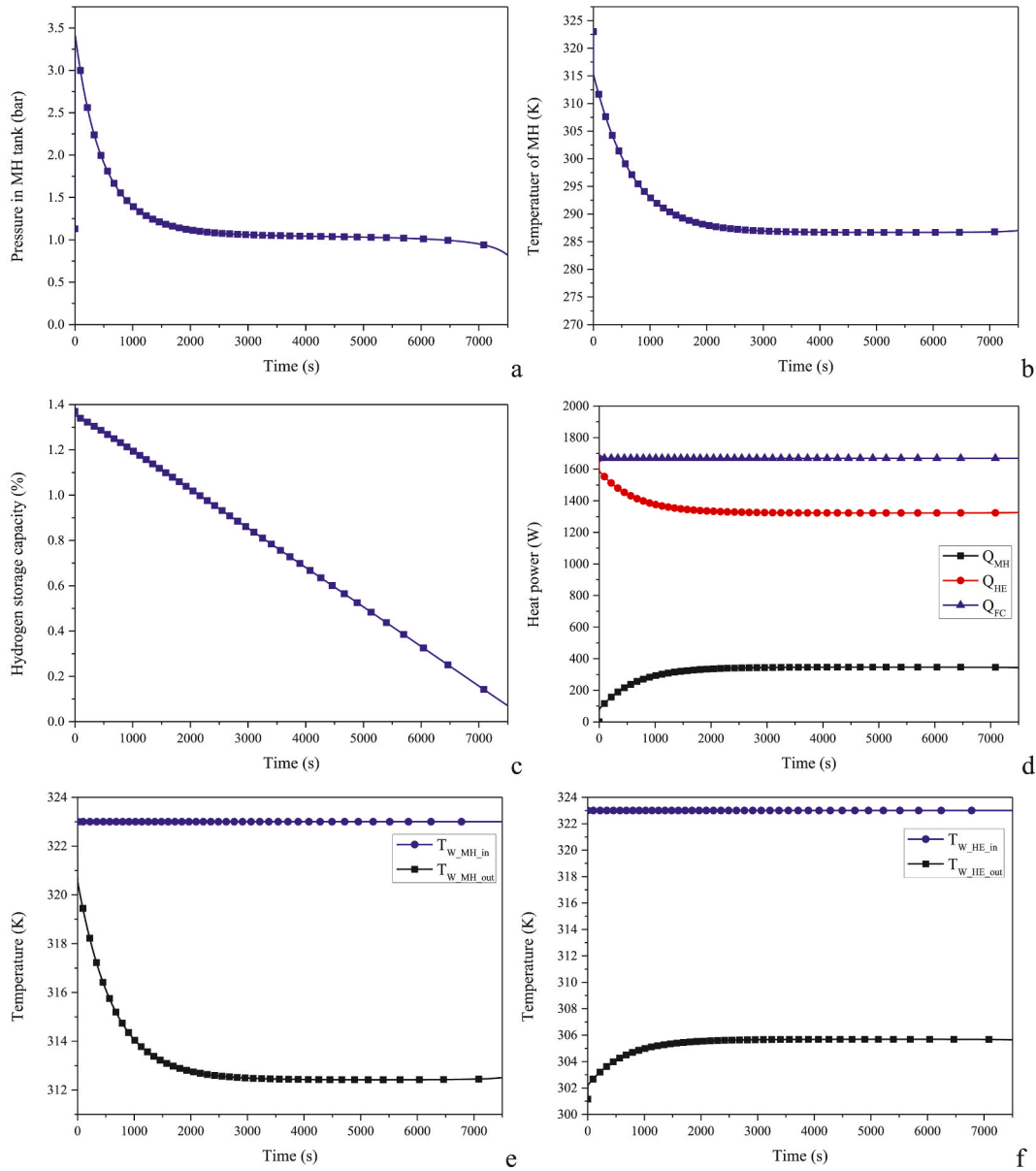


Fig. 4. Simulation results of PEMFC-MH model when the opening degree of the valve $\alpha = 0.3$ and flow rate of hydrogen from MH tank to PEMFC $\dot{m}_{H_2} = 15.17$ L/min, including (a) pressure of hydrogen in MH tank, (b) average temperature of MH bed, (c) hydrogen storage capacity of MH bed, (d) heat power, (e) temperatures at the inlet and the outlet of MH bed and (f) temperatures at the inlet and the outlet of heat exchanger.

The power of this system consisting of 45 single fuel cells, the same as that researched in the last section, is about 1 kW, which is enough to drive a light PEMFC motorcycle. As for hydrogen energy applications with relatively large power, such as the hydrogen-powered forklift in the port and the PEMFC-powered vessels on the river, multiple MH tanks can be designed and integrated as a whole to satisfy its hydrogen supply demand.

Considering the designed hydrogen supply rate of 15.17 L/min, the output power of this PEMFC can be calculated as about 926.5 W, which is almost equal to its rated power. A part of heat as the by-product of PEMFC can be used for the dehydrating reaction. The results show that the amount of hydrogen stored in this MH tank can be used for the fuel cell system to work for about 2 h.

4.2. Parametric study of PEMFC-MH system

Effects of the valve's opening degree and the flow rate of hydrogen from the MH tank to PEMFC on the performance of the PEMFC-MH system, including the pressure of hydrogen in the MH tank, the average temperature and the hydrogen storage capacity of the MH bed, the temperatures of circulating water at the outlet of the MH bed and the heat exchanger, have been carried out in this section, as shown in Figs. 5 and 6.

There are five cases with different opening degrees of the valve in Fig. 5. Considering the hydrogen supply rate of 15.17 L/min is fixed in these cases, it means the heat produced by PEMFC is constant. More heat can be distributed to the MH bed with a relatively large opening degree of the valve, which leads to increases in the gaseous hydrogen in the tank, the average temperature of the MH bed and the temperature of circulating water at the outlet of the MH bed, as shown in Fig. 5a, b and d. The effect of the opening degree of the valve on the hydrogen storage capacity of the MH bed is insignificant, as shown in Fig. 5c. The waste heat released by PEMFC is distributed to the MH bed and the heat exchanger. The more the MH bed absorbs, the less the heat exchanger exhausts. It is why the temperature of circulating water at the outlet of the heat exchanger decreases with the increase of the opening degree of the valve, as shown in Fig. 5e.

When the opening degree of the valve is set as 0.5, the pressure of gaseous hydrogen in the tank is above 1.1 bar before about 7500 s, which means that the hydrogen released from the MH tank can be directly charged into the fuel cell without additional pressurization device when the pressure loss of hydrogen in the tube is ignored. At this time, about 95% of the hydrogen stored in the MH bed

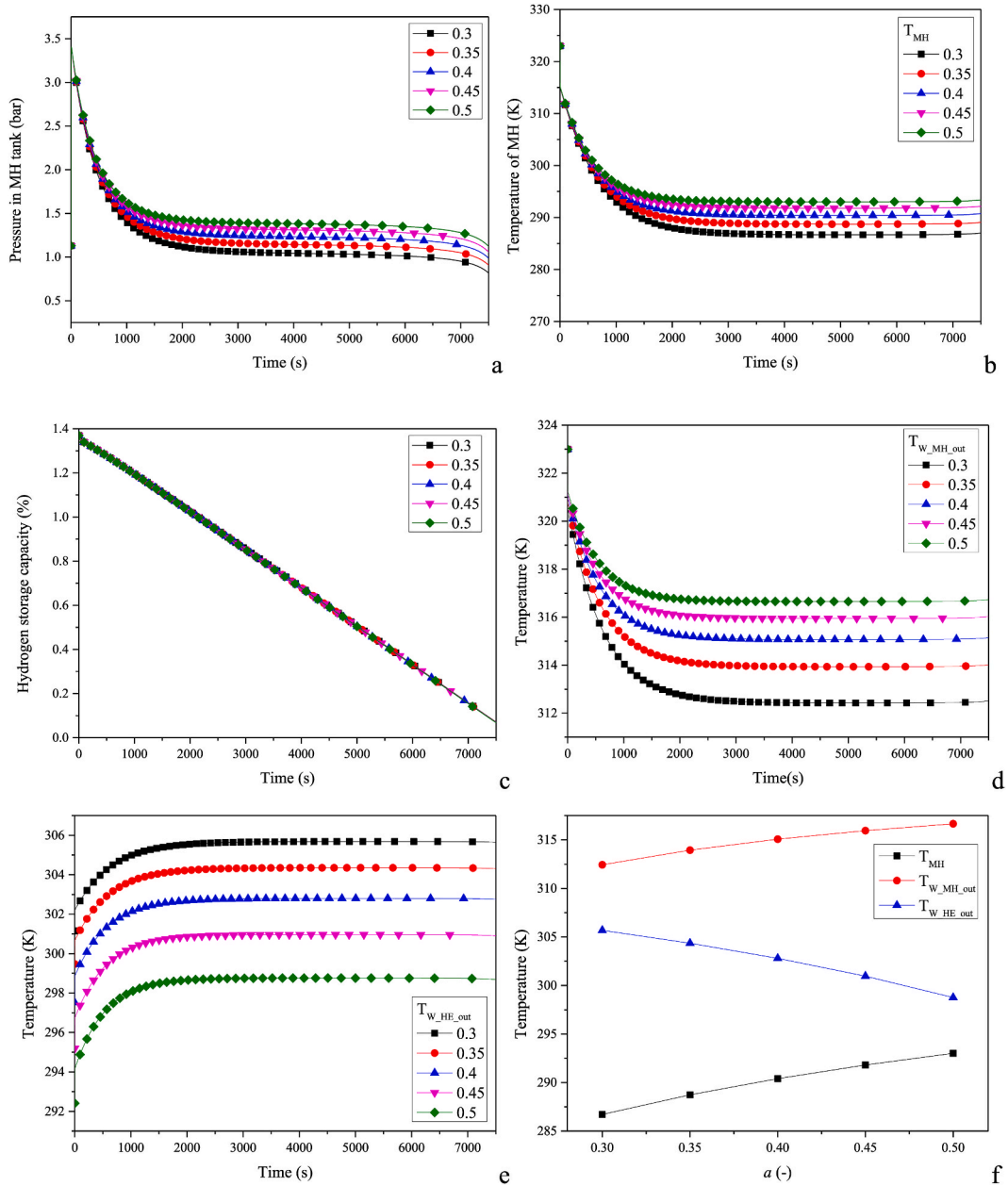


Fig. 5. Simulation results of PEMFC-MH model when the opening degree of the valve a ranges from 0.3 to 0.5 and the flow rate of hydrogen from MH tank to PEMFC is 15.17 L/min, including (a) pressure of hydrogen in MH tank, (b) average temperature of MH bed, (c) hydrogen storage capacity of MH bed, (d) temperatures at the outlet of MH bed, (e) temperatures at the outlet of heat exchanger and (f) temperatures at almost steady state ($t = 4000$ s).

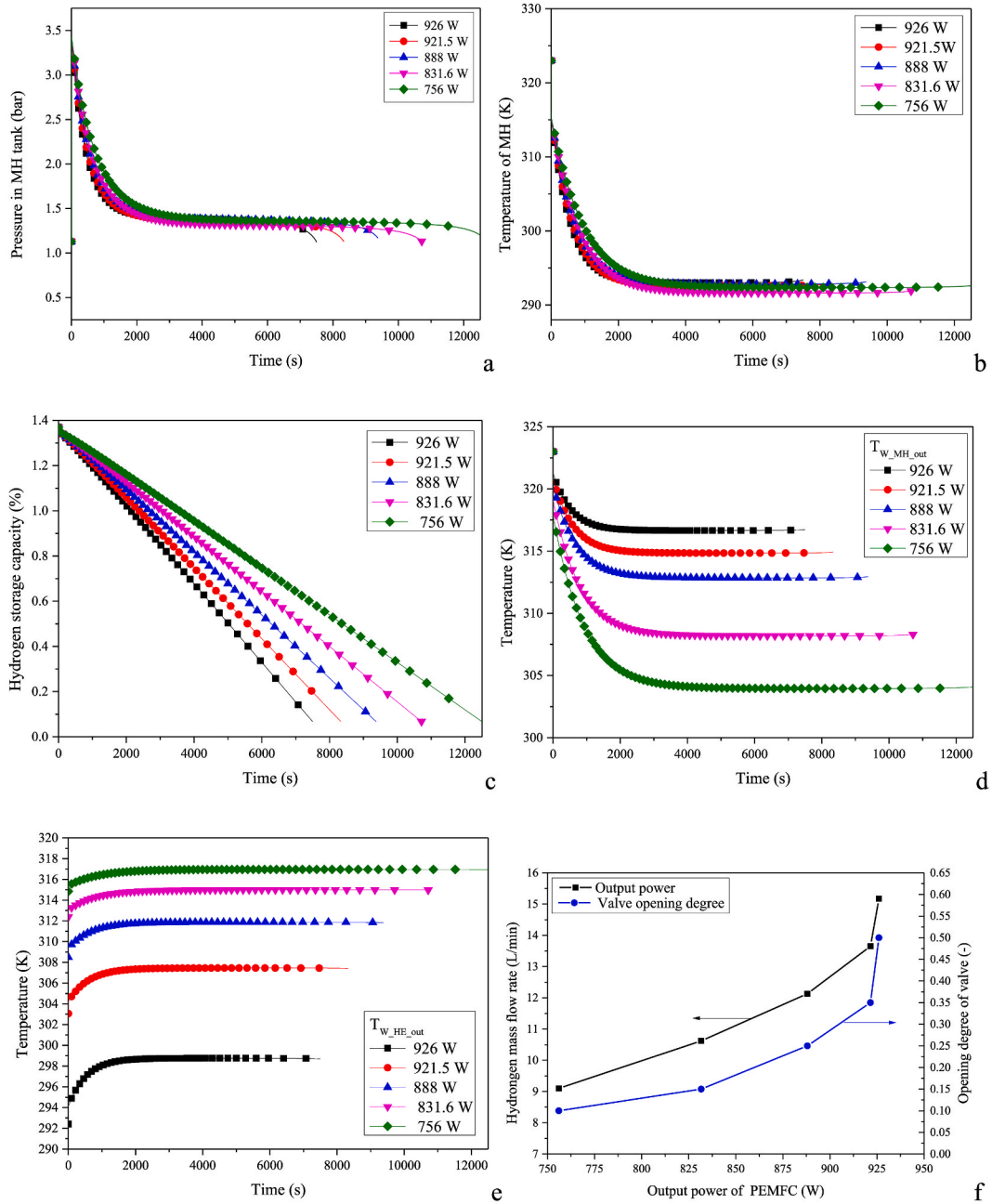


Fig. 6. Simulation results of PEMFC-MH model when the output power of PEMFC ranges from 756 W to 926 W, including (a) pressure of hydrogen in MH tank, (b) average temperature of MH bed, (c) hydrogen storage capacity of MH bed, (d) temperatures at the outlet of MH bed, (e) temperatures at the outlet of the heat exchanger and (f) hydrogen mass flow rate and the opening degree of the valve.

has been released.

The performance of the PEMFC-MH system under various output powers has been predicted in order to satisfy different operating modes for hydrogen application. The output power of PEMFC is set to 926.5 W, 921.5 W, 888 W, 831.6 W and 756.3 W, where the flow rate of hydrogen from MH tank to PEMFC respectively correspond to 15.17 L/min, 13.65 L/min, 12.13 L/min, 10.62 L/min and 9.101 L/min, as shown in Table 2. The flow rate of hydrogen in Case 5 to Case 9 decreases by 10% of that in Case 5 (about 1.517 L/min) in turn. The output power of PEMFC can be calculated by the flow rate of hydrogen. When about 95% hydrogen has been released from the MH tank and the pressure in the tank is not below 1.1 bar, the opening degree of the valve is determined. System performance with different output powers is present in Fig. 6. In general, the output power of PEMFC declines with the decrease in the flow rate of hydrogen. This MH tank with a smaller opening degree of the valve can be satisfied the hydrogen requirement for PEMFC with relatively low output power.

Table 2
Operation conditions for parametric study.

Case	Flow rate of hydrogen from MH tank to PEMFC [L/min]	Current density i_{FC} [A/cm ²]	Output power of PEMFC [W]	Opening degree of valve α [–]	Dehydrating time [s]
1	15.17	0.9648	926	0.3	7500
2	15.17	0.9648	926	0.35	7500
3	15.17	0.9648	926	0.4	7500
4	15.17	0.9648	926	0.45	7500
5	15.17	0.9648	926	0.5	7500
6	13.65	0.8684	921.5	0.35	8333
7	12.13	0.7719	888	0.25	9375
8	10.62	0.6754	831.6	0.15	10714
9	9.101	0.5789	756.3	0.1	12500

5. Conclusions

The lumped parameters model for the PEMFC-MH system composed of the PEMFC stack, the MH tank and the thermal management sub-system has been developed on the Matlab/Simulink software platform based on basic governing equations. Detailed information on the governing equations of these sub-systems has been presented in this work. The effectiveness of the model for the PEMFC stack has been validated by the data from literature. The validation of the model of a hydrogen storage tank filled with metal hydrides was done in our previous work. The model of the PEMFC-MH system with the rated output power of about 1 kW can be considered effective for studying the system performance under various conditions. A parametric study is applied to the system coupling the MH hydrogen storage tank and proton exchange membrane fuel cells to improve its energy efficiency. The opening degree of the valve and the flow rate of hydrogen have been considered in the parametric study. As for the given PEMFC-MH system, when the hydrogen supply rate is 15.17 L/min and the opening degree of the valve is set as 0.5, the MH tank can directly be used to supply the fuel to the fuel cell with the power of about 1 kW about 2 h. The hydrogen storage system can meet the hydrogen requirement of the fuel cell system when its output power ranges from 756.3 W to 926 W, where the matching opening degree of the valve has been given. Based on proper thermal management between the PEMFC stack and the MH tank, the PEMFC-MH system can present a certain degree of autonomy and stably generate electric energy without additional auxiliary heating equipment during dehydrating.

More work can be done to optimize the relationship between the capacity of the hydrogen storage tank and the output power of PEMFC. Moreover, how to effectively control the dehydrating rate when the PEMFC is started at low temperatures should be considered in the future.

CRediT authorship contribution statement

Liang Tong: writing—original draft preparation, modeling and simulation;
 Chengqing Yuan: supervision, writing—review and editing;
 Tianqi Yang: visualization, software;
 Yupeng Yuan: writing—review and editing;
 Richard Chahine: supervision, conceptualization;
 Jinsheng Xiao: methodology, writing—review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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